

1. Consider the following LP: Maximize $z = 3x_1 + 4x_2$
Subject to: $x_1 + 2x_2 \leq 4$; $2x_1 + x_2 \leq 4$; $x_1, x_2 \geq 0$
Use linprog to solve it. What is the optimal value and solution? Plot the feasible region and objective contours to verify.
2. Consider the following LP: Minimize $z = x_1 + x_2$
Subject to: $x_1 \geq 2$; $x_2 \geq 3$; $x_1 + x_2 \leq 4$; $x_1, x_2 \geq 0$
Use linprog to attempt solving it. Explain why the status indicates infeasibility, and suggest how to modify one constraint to make it feasible.
3. Consider the following LP: Minimize $z = -x_1 - x_2$
Subject to: $x_1 + x_2 \geq 5$; $x_1, x_2 \geq 0$
Use linprog to solve it. Interpret the unbounded status, and describe the direction in which the objective can decrease indefinitely. Convert it to a bounded problem by adding an appropriate constraint.
4. Consider the following LP: Maximize $z = 2x_1 + x_2$
Subject to: $x_1 + 2x_2 \leq 4$; $2x_1 + x_2 \leq 4$; $x_1, x_2 \geq 0$
Use linprog to solve it (it will return one optimal point). Analyze the problem graphically to show that there are multiple optimal solutions along an edge
5. Consider the following LP: Maximize $z = 3x_1 + 2x_2$
Subject to: $x_1 + x_2 \leq 5$; $2x_1 + 2x_2 \leq 10$; $x_1 \leq 3$; $x_1, x_2 \geq 0$
Use linprog to solve it. Note the degeneracy due to the redundant constraint. Remove the redundant constraint and re-solve to compare the outputs.